DeepFake Video Detection System for Video KYC in Financial Systems Using rPPG and Quantum Machine Learning on Low-Resolution Videos

Project Aim

This project focuses on building a deepfake video detection system for **video KYC in financial systems** where the input videos are often low in resolution and may contain compression noise, weak lighting, motion blur, and unstable face regions [web:210] [web:214]. The main goal is to detect whether a submitted KYC video is genuine or manipulated by using **rPPG-based physiological analysis as the primary evidence source** and **quantum machine learning as the final high-level decision stage** [web: 213] [web: 216] [web:219].

The proposed idea is not a generic deepfake detector for social media videos; it is designed for a financial verification setting where a wrong acceptance can support identity fraud and a wrong rejection can affect a legitimate customer journey [web:214] [web:217]. The project therefore aims to create a structured two-stage verification pipeline that works even when video quality is not ideal, which makes low-resolution robustness a core research objective rather than a side problem [web:222] [web:226].

What the Project Tries to Achieve

The project tries to achieve six practical outcomes. First, it should accept a video KYC clip and identify the face region frame by frame in a reliable manner, since all later analysis depends on consistent facial tracking [web:210] [web:215]. Second, it should extract temporal information from the facial skin region in order to estimate rPPG-style pulse-related consistency across time, because real faces preserve biological rhythm more naturally than manipulated videos [web:213] [web:216].

Third, it should classify videos as real or fake under low-resolution conditions rather than assuming clean benchmark-quality media, because real KYC environments often include compression and bandwidth limitations [web:214] [web:222]. Fourth, it should place a quantum machine learning stage after feature extraction so that the final decision is made using compact, discriminative representations rather than raw video directly [web:219][web:224]. Fifth, it should provide confidence-aware outputs that are useful for fraud screening, such as strong fake, likely fake, uncertain, likely real, and real [web:214][web:217]. Sixth, it should generate enough experimental structure for a literature review, implementation chapter, and research paper discussion [web:210][web:223].

System Flow

The overall flow of the system begins when a user uploads a short KYC video clip for identity verification [web:214][web:217]. The system then breaks the video into frames, checks frame quality, and detects the facial region in each selected frame so that only the relevant face area is passed to the analysis modules [web:210][web:215].

After face detection, the system performs alignment and normalization so that the same facial areas remain stable from frame to frame, which is very important for rPPG because pulse-related color changes are weak and can be corrupted by face jitter [web:213][web:216]. The aligned face sequence is then sent to a preprocessing pipeline that resizes the video to a fixed standard, reduces unwanted noise where necessary, and isolates skin-rich regions such as the cheeks or forehead for physiological analysis [web:216][web:226].

The first major detection stage is the **rPPG stage**, where the system studies temporal color variation patterns from facial skin regions and evaluates whether the signal behaves like a real human physiological rhythm [web:213][web:216]. The output of this stage is not just a raw label; it can include a pulse-consistency score, temporal stability indicators, and a confidence value describing whether the observed signal appears natural or suspicious [web:213][web:226].

The second major detection stage is the **quantum machine learning stage**, which receives the compact features or scores produced by the earlier pipeline and performs the final classification [web:219][web:224]. This stage acts as a higher-level judge that learns how to separate genuine and manipulated samples from the subtle patterns already extracted from the low-resolution face sequence [web:219][web:224].

Finally, the system returns a decision such as real, fake, or uncertain, along with a confidence score that can be used in a practical KYC review process [web:214][web:217]. This two-stage design is useful because the first stage focuses on biological realism and the second stage focuses on refined classification, which together create a stronger verification chain than a single shallow decision point [web:216][web:230].

Detailed Technical Architecture

The architecture can be divided into eight layers: input acquisition, frame management, face localization, low-resolution preprocessing, rPPG feature generation, feature conditioning, quantum classification, and decision support [web:210][web:213][web:219]. Each layer has a specific role and is designed so that noise is reduced gradually without destroying the weak temporal clues that matter for deepfake detection in low-quality KYC video [web:222][web:226].

1. Input Acquisition Layer

This layer accepts a raw KYC video from a user or verification interface and checks basic metadata such as file format, frame rate, duration, and video readability before further processing [web:214][web:217]. The system should reject unusable inputs such as empty files, unsupported formats, extremely short clips, or videos where a face is absent in most frames, because these cases cannot support reliable physiological analysis [web:214][web:217].

At this stage, the system can also log operational attributes such as average brightness, resolution class, and compression level for later analysis [web:222][web:223]. These values are useful during experimentation because they help explain why certain videos are easy or difficult for the model to classify [web:222][web:223].

2. Frame Sampling and Frame Quality Layer

The raw video is decomposed into frames at a controlled sampling rate rather than using every frame blindly, because stable temporal modeling often works better with a consistent and computationally manageable frame sequence [web:210][web:215]. The system should discard frames with severe blur, full occlusion, extreme head rotation, or missing facial region because such frames can distort both face-based learning and rPPG extraction [web:216][web:226].

A frame-quality unit can compute blur estimates, brightness range, and motion severity, then mark each frame as usable or weak [web:222][web:226]. This allows the system to either remove bad frames or reduce their influence when later aggregating temporal evidence [web:222][web:230].

3. Face Detection, Tracking, and Alignment Layer

The next layer detects the face box in each chosen frame and tracks that box across time to create a consistent facial sequence [web:210][web:215]. Face alignment is then applied so that major landmarks such as the eyes, nose, and mouth stay geometrically stable across frames, which reduces motion-induced signal corruption during physiological analysis [web:213][web:216].

A strong implementation should include landmark detection, similarity transform alignment, and temporal smoothing of bounding boxes to prevent sudden crop jumps [web:216][web:226]. The face crop should preserve enough skin area for rPPG while excluding excessive background, because background motion does not carry the target physiological signal and may confuse the detector [web:213][web:216].

4. Low-Resolution Preprocessing Layer

This layer standardizes all facial crops to a fixed input shape while preserving the natural temporal content of the original video [web:222][web:239]. The preprocessing should include resizing, intensity normalization, mild denoising only when necessary, and optional color stabilization so that frame-to-frame skin measurements become more reliable [web:216][web:239].

The critical design rule here is that preprocessing must **not hallucinate new detail** that did not exist in the original video, because that can distort the real forensic evidence and the weak pulse signal [web:235][web:239]. For this reason, the architecture should prefer controlled normalization over aggressive super-resolution in the main pipeline [web:235][web:237].

At a minute-detail level, this layer can include: fixed-size face resizing; temporal smoothing for minor jitter; histogram or brightness normalization; ROI masking for cheeks and forehead; and per-frame quality tagging [web:216][web:226]. Each of these micro-steps serves one purpose: keep the signal consistent enough for rPPG analysis without cosmetically rewriting the content [web:213][web:216].

5. rPPG Feature Generation Layer

This is the **core hero layer** of the project because it extracts physiological evidence that is difficult for many deepfakes to preserve [web:213][web:225]. The layer first selects skin-rich regions, such as left cheek, right cheek, and forehead, and then tracks their color intensity changes over time in order to estimate a pulse-related signal [web:213][web:216].

The model can derive temporal descriptors such as periodicity, inter-region consistency, waveform stability, amplitude reliability, and synchronization behavior across facial areas [web:213][web:226]. Real videos are expected to show more coherent and biologically plausible temporal patterns, while fake videos may show weak, unstable, or inconsistent pulse-like behavior because generative manipulation often focuses on appearance realism rather than true physiological continuity [web:213][web:216].

A detailed implementation can include: ROI extraction, temporal signal buffering, channel-wise color aggregation, detrending, band-related filtering, normalization of the recovered signal, and transformation of the resulting sequence into compact features or a short temporal embedding [web:216][web:225]. The output of this layer should not be only a binary tag; it should be a structured physiological feature representation that captures whether the face behaves like a living face over time [web:213][web:216].

6. Feature Conditioning and Fusion Layer

Before entering the quantum stage, the architecture should convert the rPPG outputs and optional auxiliary low-level video statistics into a compact, normalized feature vector [web:219][web:224]. This step is important because quantum machine learning models are generally more suitable for small and meaningful feature spaces than for raw high-dimensional video tensors [web:219][web:224].

The conditioning unit can include feature normalization, dimensionality reduction, confidence calibration, and concatenation of support features such as frame quality score, ROI stability score, or temporal consistency score [web:222][web:230]. In this design, the quantum stage does not replace physiological extraction; instead, it receives distilled evidence from the earlier layers [web:219][web:224].

7. Quantum Machine Learning Classification Layer

This layer acts as the **second hero layer** of the project because it performs the final discrimination using the compact evidence produced earlier [web:219][web:224]. The intuition is that low-resolution deepfake detection produces subtle, noisy, hard-to-separate patterns, and a quantum classifier can be explored as a novel decision engine for those compact patterns rather than for raw video pixels [web:219][web:224].

In practical system design terms, the conditioned feature vector is mapped into the quantum model, where it is processed by a hybrid classical-quantum classification block and then converted into a final prediction score [web:219][web:224]. The output can include fake probability, real probability, and uncertainty level, which supports both experimental evaluation and applied KYC screening [web:214][web:217].

A safe project implementation should keep this as a **hybrid** stage rather than a purely standalone quantum system, because the realistic strength of the project comes from combining classical preprocessing and physiological extraction with a novel quantum decision mechanism [web:219][web:224]. This also makes the pipeline more understandable, more trainable, and easier to explain in a research paper [web:219][web:223].

8. Decision and Review Layer

The last layer converts the classifier outputs into a usable financial-verification result [web:214][web:217]. Instead of only saying fake or real, the system should support confidence-aware decisions such as accept, reject, or manual review required [web:214][web:217].

This is especially important in a KYC setting because uncertain cases should not be forced into a risky binary answer [web:214][web:217]. A review-oriented design also makes the project more practical and aligns it with fraud prevention workflows [web:212][web:214].

Why rPPG and Quantum Machine Learning Are the Heroes

rPPG is the first hero because it gives the project a **biological basis** for detection rather than relying only on visible artifacts [web:213][web:225]. Deepfakes can often improve visual appearance quickly, but preserving realistic pulse-related facial color dynamics across time is much harder, especially in manipulated or synthesized videos [web:213][web:216].

This matters even more in your project because the videos are low resolution, which weakens fine texture analysis and makes ordinary image-forensics cues less dependable [web:222][web:226]. rPPG shifts the focus from “does the face texture look fake?” to “does this face behave like a live human face over time?” which is a stronger identity-verification perspective for video KYC [web:213][web:214].

Quantum machine learning is the second hero because it gives the project its **novel decision intelligence layer** [web:219][web:224]. After the rPPG stage extracts subtle temporal-biological clues, the quantum layer becomes responsible for learning how those clues separate real and fake samples under noisy low-resolution conditions [web:219][web:224].

In other words, rPPG produces the evidence and quantum machine learning judges that evidence [web:213][web:219]. This makes both components central to the identity of the project: one is the physiological truth source, and the other is the research-level classification engine [web:213][web:224].

Low-Resolution Video Preprocessing and Why Not Full Super-Resolution

Low-resolution KYC videos should absolutely be preprocessed, but the preprocessing should aim for **consistency**, not cosmetic enhancement [web:222][web:239]. The best mainline pipeline is usually face cropping, alignment, resize normalization, mild denoising if needed, and stable ROI extraction rather than heavy visual beautification [web:216][web:239].

The reason full super-resolution is not ideal as the default approach is that it can invent visual detail that was never present in the original video, and those artificial details may interfere with forensic cues and physiological estimation [web:235][web:239]. For a project whose main evidence depends on weak temporal facial patterns, preserving original signal behavior is more important than making the face look sharper to the human eye [web:213][web:239].

A careful preprocessing policy should therefore do the following: standardize frame size, stabilize face crops, normalize brightness, isolate skin regions, remove clearly unusable frames, and preserve the original temporal ordering of the signal [web:216][web:226]. Optional super-resolution can still be tested as a side experiment for comparison, but it should not define the core architecture unless results clearly prove that it improves physiological detection without damaging signal integrity [web:235][web:237].

Recommended Final Working Plan

The most practical working plan for the project is to implement the system as a **two-stage pipeline** [web:216][web:230]. Stage 1 performs low-resolution face preprocessing and rPPG-based physiological analysis, while Stage 2 performs hybrid quantum-based classification on the extracted compact features [web:213][web:219].

This gives the project a clear identity, strong novelty, and a clean research story for the final paper: financial video KYC deepfake detection in low-resolution settings using rPPG as the main evidence source and quantum machine learning as the final decision engine [web:214][web:219][web:223].